

The slope formula



- 1

a 3	b -3	c -1	d -1
e $\frac{3}{5}$	f -1	g 0	h 0

- 2 No. Reversing the order does not affect the final outcome of the slope. As long as you maintain the correct order by subtracting the corresponding y and x coordinates, the slope should be the same.

- 3

a Undefined	b Undefined	c Defined	d Defined
e Undefined	f Defined	g Undefined	h Defined

- 4

a $y = 25$	b 5
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- 5

a $t = 3$	b $t = 15$	c $d = 20$	d $c = -\frac{12}{7}$
e $t = -23$			

- 6 890 m

- 7

a 0	b 4	c No
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- 8

a undefined	b No	c 1
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- 9

a Neither	b Undefined	c 0	d Neither
e Neither	f Neither	g 0	

- 10

a The change in x is equal to zero. The line is vertical.	b The line is horizontal. The change in y is equal to zero.
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11

- a No
- b Yes
- c No

12 D

13

- a 0
- b Undefined

14 B

15 -20

Additional questions

16

- a
 - i 14
 - ii 7
 - iii 2
- b
 - i 12
 - ii 4
 - iii 3
- c
 - i -12
 - ii 4
 - iii -3
- d
 - i -6
 - ii 2
 - iii -3

17 Because a vertical line has no run. So the formula for the slope $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a zero denominator is undefined.

18 0

19

- a 2
- b $-\frac{7}{5}$
- c $\frac{1}{2}$
- d $-\frac{2}{3}$

20

a -2



b As x increases, the value of y decreases.

21

a 3

b $\frac{3}{5}$

c -4

d -3

e 0

f 0

g 1

h -1

22

a $\frac{5}{3}$

b $\frac{5}{3}$

c Yes

23 $m = 1$